

22.10.22

MID TERM EXAMINATION 2022-23

TPE-511

MACHINE LEARNING FOR RESERVOIR CHARACTERIZATION

(Petroleum Engineering)

Maximum Marks: 50

Duration: -1:30 hours

NOTE:

- All questions are mandatory.
- Any two out of three choices should be attempted in each question.

Q. 1.) a.) State the difference between Artificial Intelligence, Machine Learning and Deep learning. [CO3]

b.) What are three different Machine Learning types. Explain each. [CO1]

c.) Explain the Following: [CO2]

i.) Problem Solving Approach

ii.) Execution Time

iii.) Interpretability

Q.2.) a.) Explain any two of the Following Algorithms: [CO3]

i.) Classification Algorithm

ii.) Anomaly Detection Algorithm

iii.) Clustering Algorithm

iv.) Regression Algorithm

b.) What is Data? Explain different categories of Data. [CO2]

c.) Explain basic terminologies in Statistics. [CO2]

Q.3.) a.) With a neat diagram explain Sampling Techniques. [CO2]

b.) State the difference between Descriptive Statistics and Inferential Statistics. [CO2]

c.) State the relationship between Variance and Deviation. Give examples. [CO2]

Q.4.) a.) Explain how biased sample population is used as the conditioning data during construction of a geostatistical reservoir model? [CO1]

b.) Explain the four measurement scales. [CO2]

c.) What is EDA? [CO2]

Q.5.) a.) A scientist is researching whether or not birds of prey exposed to pollutants lay eggs with thinner shells. He collects a random sample of egg shells from each of 6 different nests and tests for pollutant level, p , and measures the thinning of the shell, t . The results are shown in the table below. [CO4]

p	3	8	30	25	15	12
t	1	3	9	10	5	6

(You may use $\sum p^2=1967$ and $\sum pt= 694$)

- I. Draw a scatter diagram to represent these data.
- II. Explain why linear regression model may be appropriate to describe the relationship between p and t .
- III. Calculate the value of S_{pt} and S_{pp} .
- IV. Find the equation of the regression line of t on p , giving your answer in the form $t = a + bp$.
- V. Plot the point $(\bar{p}, \bar{t})$ and draw the regression line on your scatter diagram.

b.) Apply K-means algorithm to partition the below dataset into two clusters. Take the initial centroid of two clusters as (1, 1) and (2, 4). Use Euclidian distance metric. Perform up to maximum of one relocation. Show your work step by step. [CO4]

Subject	A	B
1	1	1
2	1	2
3	1	4
4	2	4
5	2	5
6	1	3

c.) Explain the working principle of "Random Forest Algorithm". With Example. [CO4]